Day 4 Report — Banking Intent Classifier

October 14, 2026

Summary

TF-IDF + Logistic Regression is our final model. On the test set it reaches a Macro-F1 of 0.886, against 0.834 for the LSTM and 0.0003 for the Dummy baseline.

Day 4 was for finishing, not for new experiments. What we did:

- **Final model selection.** We compared every model on validation and checked the four questions of the guide: performance, simplicity, interpretability and stability.
- **Final evaluation.** We evaluated the test set at the end. The scores are identical to Day 2, so the final model did not change.
- **Final error analysis.** We counted where each model fails and which intents are confused, and we listed the words the model relies on.
- **Reproducibility check.** Retraining from scratch gives identical predictions, and the saved pipeline matches the code.
- **Final pipeline.** The final models are saved in a new folder, `saved_model_final/`, so the Day 3 version stays untouched.
- **Presentation.** The final deck has 5 slides, with one part for each team member and short speaking notes.

Rules for Today

We only finished what already worked. The task, the data and the model direction did not change.

Rule	How we applied it
Work on a copy	We made a backup copy and a final copy of the Day 3 notebook, and worked only in the final copy
Keep the working pipeline safe	The final models are saved to `saved_model_final/`. The Day 3 `saved_model/` folder and the saved_model.zip that we downloaded stay unchanged
Same text preparation	The same tokenizer, vocabulary (2,205 tokens), maximum length (33) and label mapping as Day 3

Rule	How we applied it
Validation chooses	The final model was chosen on validation and cross-validation only
Test reports	The test set was evaluated at the end. We had also seen the same scores on Day 2, so we did not use them to choose or tune any model
No new direction	No new models, no new data, no new preprocessing

Final Model Selection

TF-IDF + Logistic Regression has the highest validation score, about 5 points above every LSTM version.

All models on validation

Model	Accuracy	Macro-F1	Top-3
TF-IDF + LogReg (tuned)	0.8861	0.8870	0.9690
TF-IDF + LogReg	0.8861	0.8869	0.9680
LSTM, Exp 2: dropout 0.5	0.8361	0.8359	0.9375
LSTM, Exp 1: max length 48	0.8311	0.8310	0.9345
LSTM pipeline (Day 2)	0.8301	0.8296	0.9305
Dummy (most frequent)	0.0190	0.0005	0.0490

Tuning check

We checked the settings of TF-IDF + Logistic Regression with a grid search: C in {3, 10, 30} and word ranges (1, 2) or (1, 3), with 5-fold cross-validation on the training split only. The best setting was C = 30 with words and word pairs (cross-validation Macro-F1 0.8738, against 0.8710 for the current model). On validation it gives 0.8870, against 0.8869.

The gain is 0.0001, which is noise. We kept the original model (C = 10), so the saved pipeline, the reports and the demo stay the same.

Why this model?

Question	TF-IDF + Logistic Regression	LSTM
Performance	Highest Macro-F1: 0.887 on validation, 0.886 on test	0.830 to 0.836 on validation, 0.834 on test
Simplicity	One vectorizer and one linear classifier. Trains in seconds on a CPU	347,681 parameters, many epochs, early stopping
Interpretability	Every intent has readable word weights (see Final Error Analysis)	A black box
Stability	5-fold CV Macro-F1 0.871 ± 0.006. Validation 0.887 and test 0.886 agree	Overfits: 97.2% training vs 82.5% validation accuracy

Final Evaluation (Test Set)

The test set (3,080 messages, 40 per intent) was evaluated at the end, after the final model was chosen. We had also seen the same scores on Day 2, so we did not use them to choose or tune any model.

Model	Accuracy	Macro-F1	Weighted-F1	Top-3
TF-IDF + Logistic Regression	0.8854	0.8858	0.8858	0.9685
LSTM pipeline	0.8344	0.8340	0.8340	0.9481
Dummy (most frequent)	0.0130	0.0003	0.0003	0.0390

The scores are identical to the Day 2 test run, which confirms that the final model is the same one. The validation score (0.8869) and the test score (0.8858) are very close, so the model does not look over-tuned on validation. TF-IDF is 5.2 points of Macro-F1 above the LSTM.

Final Error Analysis

TF-IDF makes fewer errors than the LSTM, and its errors are spread over many intents rather than concentrated in a few.

Validation (2,001 messages)	Count
TF-IDF errors	228 (11.4%)
LSTM errors	340 (17.0%)
Both models wrong	151
LSTM wrong, TF-IDF right	189
TF-IDF wrong, LSTM right	77

The 10 intents with the most TF-IDF errors hold only 31% of all its errors.

Short and long messages

The LSTM loses about 10 points on long messages, while TF-IDF stays stable. Accuracy on validation:

Message length	Messages	TF-IDF	LSTM
Short (up to 8 tokens)	500	85.0%	82.8%
Medium (9 to 13)	936	90.8%	86.3%
Long (14 to 20)	328	87.8%	77.7%
Very long (over 20)	237	88.6%	77.6%

Most confused intents (TF-IDF, validation)

The model confuses intents that mean almost the same thing. Count = number of validation messages.

True intent	Predicted intent	Count
direct_debit_payment_not_recognised	card_payment_not_recognised	6
get_disposable_virtual_card	disposable_card_limits	4
declined_cash_withdrawal	declined_transfer	3
why_verify_identity	unable_to_verify_identity	3
change_pin	get_physical_card	3
country_support	order_physical_card	3

True intent	Predicted intent	Count
top_up_failed	pending_top_up	3
balance_not_updated_after_bank_transfer	pending_transfer	3

Important features

The words with the largest weights for each intent make sense to a person, so every prediction can be explained.

Intent	Top words and word pairs
card_arrival	card, track, my new, tracking, new card, my card
lost_or_stolen_card	stolen, lost, my card, card, lost my, i lost
declined_transfer	declined, transfer, my transfer, declined ., decline, message
exchange_rate	exchange, exchange rate, rate, rates, exchange rates, your exchange
top_up_failed	up, top, my top, top up, failed, worked
pending_top_up	up, pending, my top, top, top up, through

The last two intents share most of their top words ("top up", "my top"). This may explain why top-up intents are often confused.

Limitations

The model always picks one of the 77 intents, even for messages that have nothing to do with banking. We tested it with three such messages:

Message	Predicted intent	Confidence
"What is the weather today?"	card_not_working	8%
"I want to order a pizza"	request_refund	25%
"How do I bake a cake?"	transfer_into_account	12%

The confidence is low in all three cases (8% to 25%), so a confidence threshold could send such messages to a person.

Other limitations:

- **Small data.** 8,002 training messages for 77 intents, about 100 per intent and only 28 for the smallest one.
- **Close intents.** Some intents mean almost the same thing, and some labels are ambiguous, so part of the error comes from the labels.
- **Words, not context.** TF-IDF counts words and word pairs. It does not understand the meaning of a full sentence.
- **Clean English only.** Banking77 is clean, written English. Real customer chat has more typos, slang and other languages.
- **The LSTM overfits.** It reaches 97.2% on training but 82.5% on validation, and it is weaker on long messages.

Reproducibility Check and Final Pipeline

The final pipeline is reproducible: retraining gives identical predictions, and the saved files match the code.

Check	Result
Retrain TF-IDF + Logistic Regression from scratch	Identical predictions on all 2,001 validation messages
Vocabulary and maximum length on disk	Match the text preparation of the saved LSTM (2,205 tokens, 33)
Class names on disk	Match the label encoder (77 intents, same order)
Reload the saved models	Same predictions as the models in memory (checked on 200 validation messages)
Test scores	Identical to the Day 2 test run

Saved files (`saved_model_final/`)

File	Contents
`tfidf_logreg.joblib`	The final model: TF-IDF vectorizer and Logistic Regression in one pipeline
`lstm_model.keras`	The LSTM, kept as the deep-learning comparison
`vocab.json`	The LSTM vocabulary and the maximum length

File	Contents
`classes.json`	The 77 intent names in label-encoder order

The folder is downloaded as `saved_model_final.zip`. The Day 3 `saved_model.zip` that we downloaded is kept unchanged, so we can always go back to the version that worked yesterday.

Final Visualizations and Presentation

The final figures are saved as PNG files for the slides, and the deck tells the full story in 5 slides.

Final figures (`final_figures/`)

File	What it shows
`model_comparison.png`	Validation Macro-F1 of every model, from the Dummy to TF-IDF
`accuracy_by_length.png`	Accuracy of TF-IDF and the LSTM by message length
`top_confusions.png`	The 8 most confused intent pairs for TF-IDF

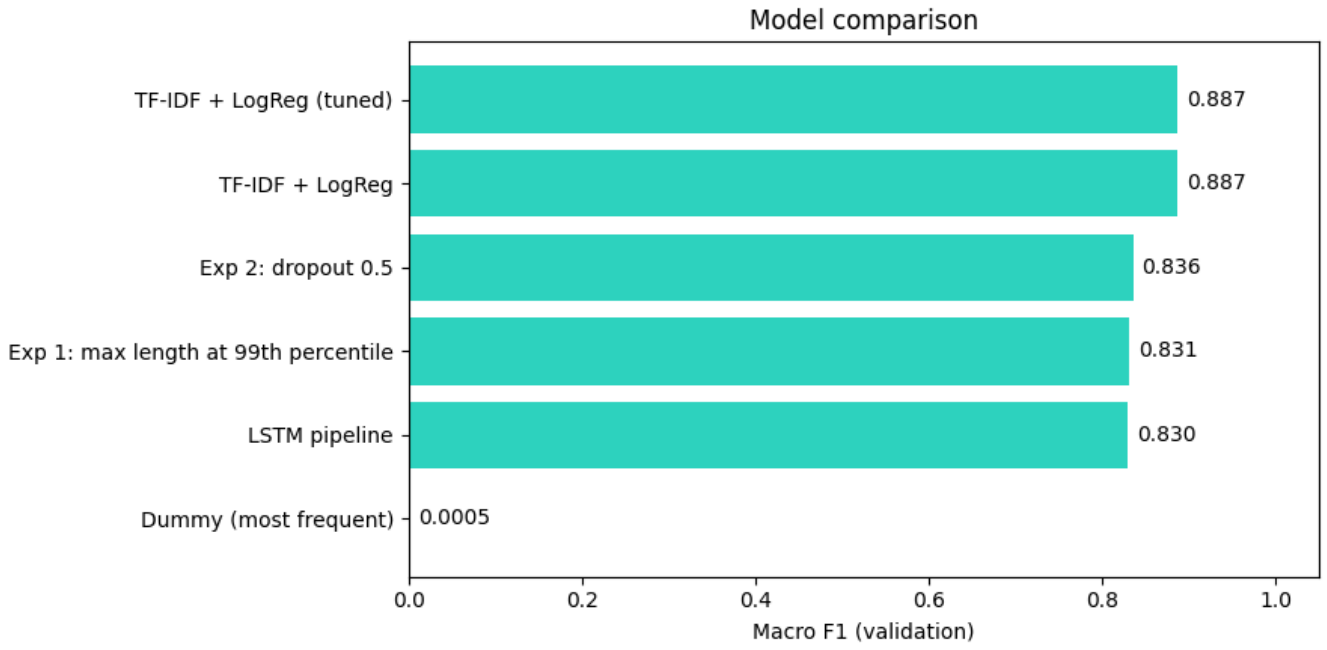


Figure 1. Validation Macro-F1 of every model. TF-IDF + Logistic Regression (0.887) is clearly above all LSTM versions (0.830 to 0.836).

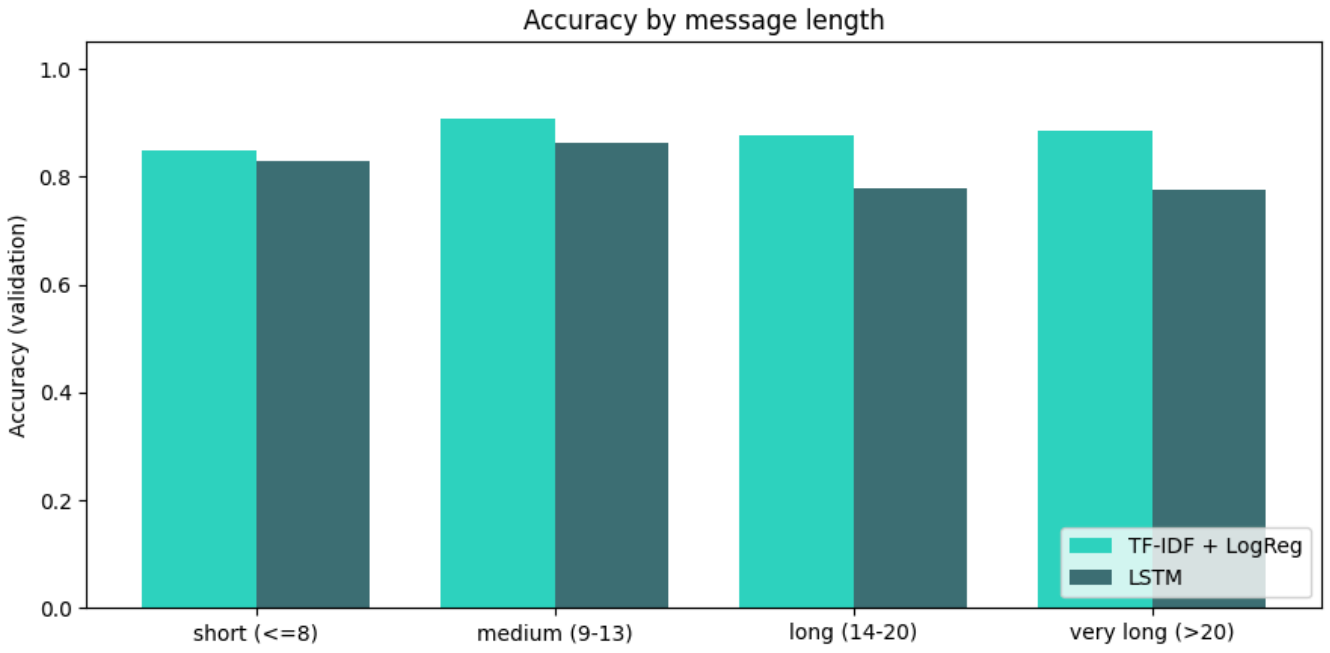


Figure 2. Validation accuracy by message length. TF-IDF stays stable, while the LSTM drops on long messages.

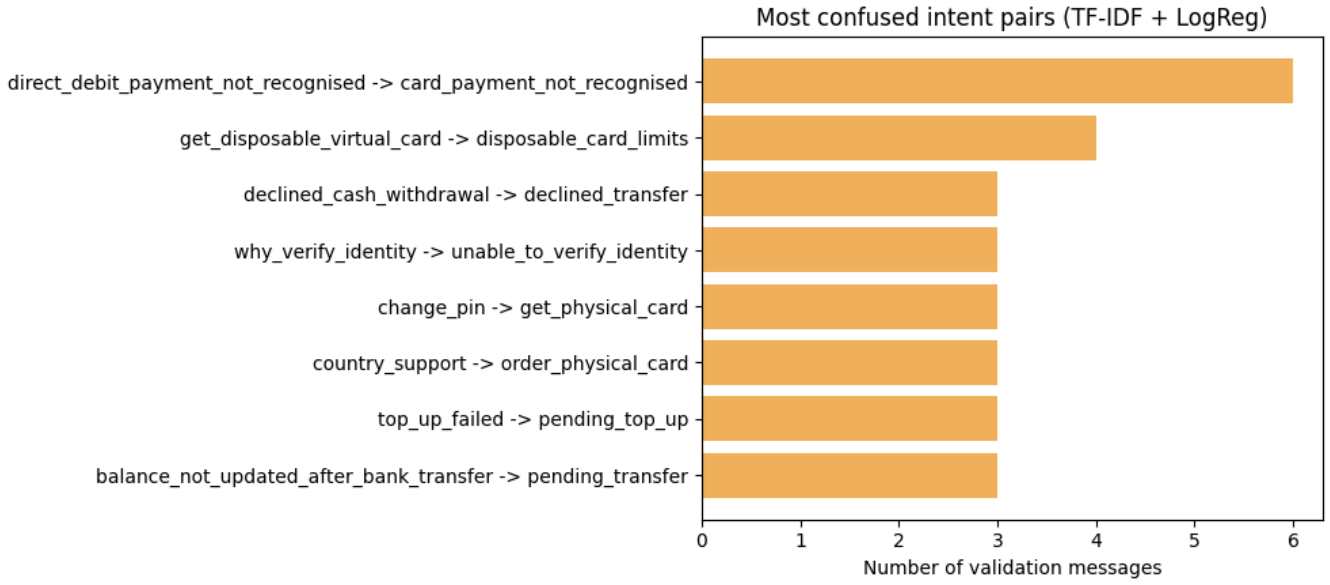


Figure 3. The 8 most confused intent pairs for TF-IDF on validation. All of them are intents with close meanings.

Final presentation

#	Slide	Main message
1	Title and team	A model that reads one short banking message and predicts what the customer wants, out of 77 intents

#	Slide	Main message
2	The problem	Banks receive thousands of messages, and each one needs the right team. A lost card must never be mixed up with a harmless question
3	The data	77 intents, 8,002 training, 2,001 validation and 3,080 test messages. Short messages, close intents, light text preparation fitted on train only
4	The approach	Dummy, TF-IDF + Logistic Regression and an LSTM, then two controlled LSTM experiments
5	Results	TF-IDF wins: 0.887 on validation and 0.886 on test. Live demo and conclusion

Each of the four team members presents one part, with short speaking notes in simple English: main points, important numbers and a transition sentence.

Demo. The final slide ends with a live demo in our Streamlit app (see the next section). The slides also explain the results without the demo, in case it cannot run.

Demo App (Streamlit)

The demo is a Streamlit web app (`app.py`) that runs on our laptop (localhost). It reads the saved models from a folder named `saved_model/` next to app.py, so no training happens in the app. This folder has the same files as saved_model_final/, because the final model did not change.

What the user sees

Part of the app	What it does
Title and intro	"Type a customer message. The saved model tells us what the customer wants (one of 77 intents). No training happens here."
Example buttons	Lost card, Declined transfer, Top-up time and Card not arrived. One click fills in a prepared message
Customer message box	Any message can be typed, for example "I lost my card yesterday, please block it"
Predict button	Runs the saved TF-IDF + Logistic Regression model on the message

Part of the app	What it does
Prediction	The predicted intent in readable words and its confidence
Top 3	The three most likely intents, with a bar for each probability
Low-confidence message	When the confidence is below 50%: "The model is not very sure. A person should check this message."
Sidebar: About the models	TF-IDF + Logistic Regression is the main model (test Macro-F1 0.886). The LSTM is the comparison (test Macro-F1 0.834). Trained on PolyAI Banking77: 77 intents, about 10,000 messages
Sidebar checkbox	"Also show the LSTM (slower to load)". When it is on, the app also shows the LSTM top 3 and says when the two models disagree

How it works

1. **Load once.** The app loads the TF-IDF + Logistic Regression pipeline, the 77 class names and the LSTM vocabulary from disk, and keeps them in memory.
2. **Same text preparation.** It uses the same `clean_text` and `tokenize` functions as the training notebook. The saved pipeline needs them, and a different preparation would give different predictions.
3. **Predict.** The pipeline returns a probability for each of the 77 intents. The app shows the highest one and the top 3.
4. **Optional LSTM.** The LSTM needs TensorFlow, which is slow to load, so it is loaded only when the checkbox is on. If it cannot be loaded, the app shows TF-IDF only.

Results of the example buttons

Button	Message	Predicted intent	Confidence
Lost card	I lost my card yesterday, please block it	lost or stolen card	66%
Declined transfer	Why was my transfer declined?	declined transfer	88%
Top-up time	How long does a top up take?	pending top up	52%

Button	Message	Predicted intent	Confidence
Card not arrived	My card has not arrived yet	card arrival	44%

"Card not arrived" is below 50%, so the app also shows the low-confidence message. Its second guess is transfer_not_received_by_recipient (20%).

The app uses one threshold: when the confidence is below 0.5, it asks a person to check the message. This way a bank can answer clear messages automatically and send uncertain ones to a person.

If the live demo cannot run

The slides show the demo results, and the prediction notebook gives the same predictions from the same saved model. The presentation does not depend on the live app.

Day 4 Checklist

The project and the code are finished. The presentation is complete and ready for the final rehearsal.

Area	Item	Status
Project	Final model selected	Done: TF-IDF + Logistic Regression
Project	Final results saved	Done: final scores in the notebook and in this report, figures in `final_figures/`
Project	Metrics checked	Done: Macro-F1 on validation, cross-validation and test
Project	Error analysis completed	Done: errors per model, length, confused pairs, important features
Project	Limitations identified	Done
Code	Important code works	Done: every cell runs without errors
Code	Final notebook or demo prepared	Done: final notebook, prediction notebook and Streamlit app
Presentation	Slides complete	Done: 5 slides
Presentation	Everyone has a speaking part	Done: one part per team member
Presentation	Speaking notes prepared	Done

Area	Item	Status
Presentation	Difficult English simplified	Done: short sentences and simple words
Presentation	One complete rehearsal, timing checked	Planned for the end of Day 4, with the real slides and a timer

Pitfall	How we avoided it
Starting a new experiment	No new models, data or preprocessing. The tuning check only confirmed the existing model
Tuning on the test set	We did not use the test scores to choose or tune any model
Leaving the slides to the last minute	The slides are complete, with no placeholders for important results
Depending on a live demo	The slides explain the results without the demo